

Injury Data Analysis

Final Case Project

Introduction



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Data Analytics
Fifth-Third Bank

Top 3 Injuries

Most expensive injuries, based on average cost:

- Multiple Injuries
- Dislocation
- Tear

Most expensive injury is over 221 times greater than least expensive.

Each top 3 injury is about two times more expensive than the average (\$15,247.49).

Top 3 injuries make up 7% of total injury cost.

Source: Generic Utility Company Injury Data n = 654

Top Injuries by Average Cost



Nature of Injury	
Multiple Injuries	\$34,792.59
Dislocation	\$32,778.22
Tear	\$28,941.58
Sprains/Strains	\$25,486.35
Inflammation of Joints	\$18,705.77
Concussion	\$18,681.64
Hernia- Rupture	\$16,942.42
Non-Classifiable	\$16,721.17
Fracture	\$15,709.52
Crushing	\$12,215.44
Contusions	\$9,433.41
Abrasion	\$7,315.77
Burn	\$3,739.78
Infection/Inflammation	\$3,249.24
Laceration	\$2,387.72
Heat Stress	\$1,760.37
Electrocution	\$1,235.85
Cumulative Trauma Disorder	\$1,058.84
Puncture	\$791.99
Hearing Loss or Impairment	\$735.97
Animal/Insect bite	\$492.64
Poisoning	\$444.86
Skin Rash	\$237.76
Asphyxiation	\$157.08

Body Parts Affected by Top Injuries

Injury type sorted by highest average cost, highest to lowest.

Included Sprains/Strains (top 4 injury) due to Multiple Injuries/Body Parts being vague.

Shoulder appears in every category.

Knees appear in all but one category.

Source: Generic Utility Company Injury Data n = 654

Top Body Parts Injured

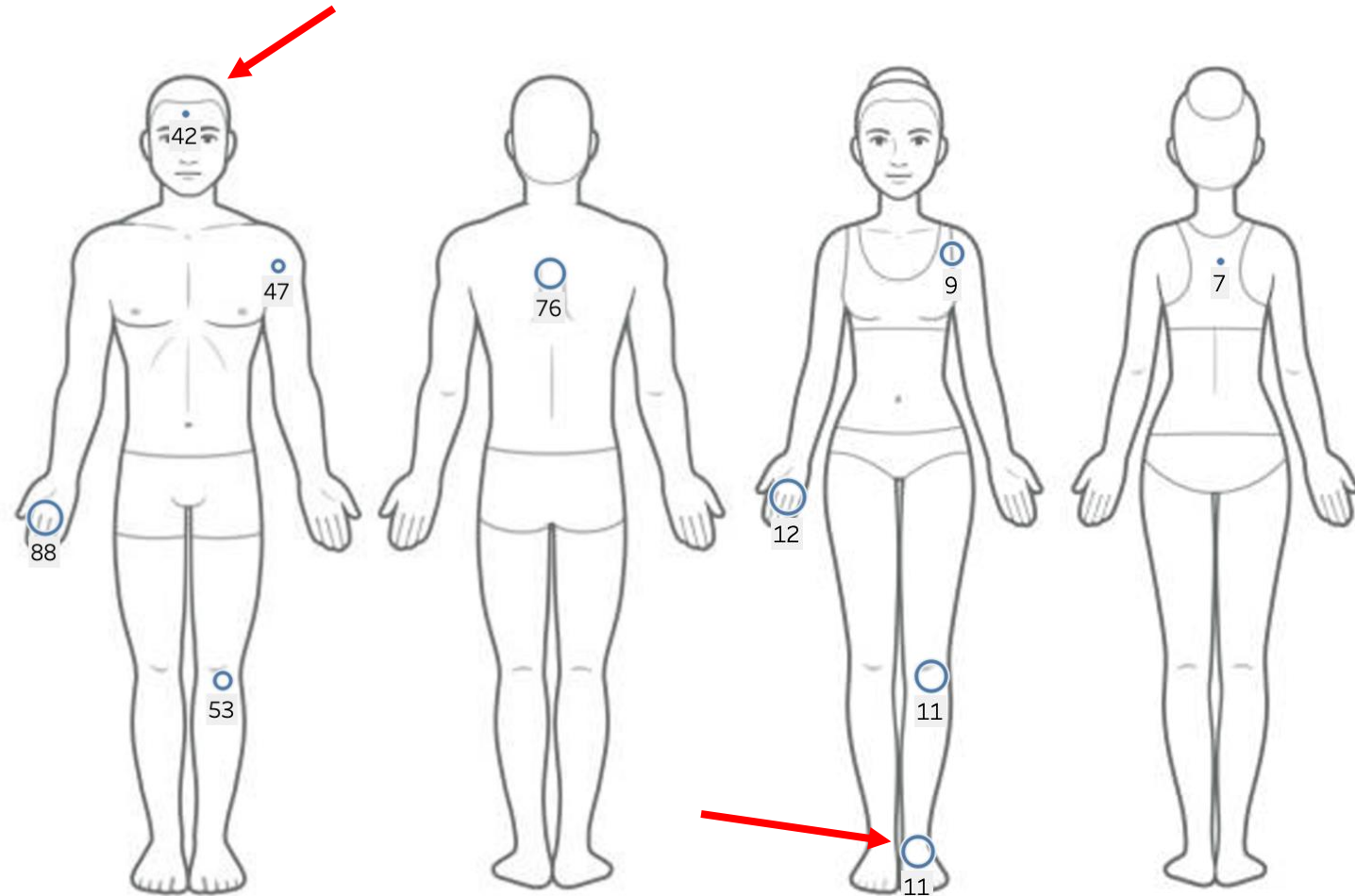
Nature of Injury	Part of Body Affected	
Multiple Injuries	Multiple Body Parts	5
	Shoulder	1
	Knees	1
	Hip/Buttock	1
	Back	1
Dislocation	Shoulder	2
	Hand	2
Tear	Knees	3
	Legs	2
	Arm	2
	Shoulder	1
	Elbow	1
Sprains/Strains	Back	60
	Shoulder	43
	Knees	39
	Ankle	27
	Neck	13
	Wrist	8
	Legs	7
	Hand	7
	Arm	6
	Multiple Body Parts	5
	Hip/Buttock	4
	Feet	4
	Elbow	4
	Ribs	3
	Groin	3
	Abdomen	3
	Head	2
	Chest	2

Most Injured Body Parts for Men and Women

Diagram shows top 5 most injured body parts for men and women.
Sizing is relative within gender.

Most injuries occur on hands.

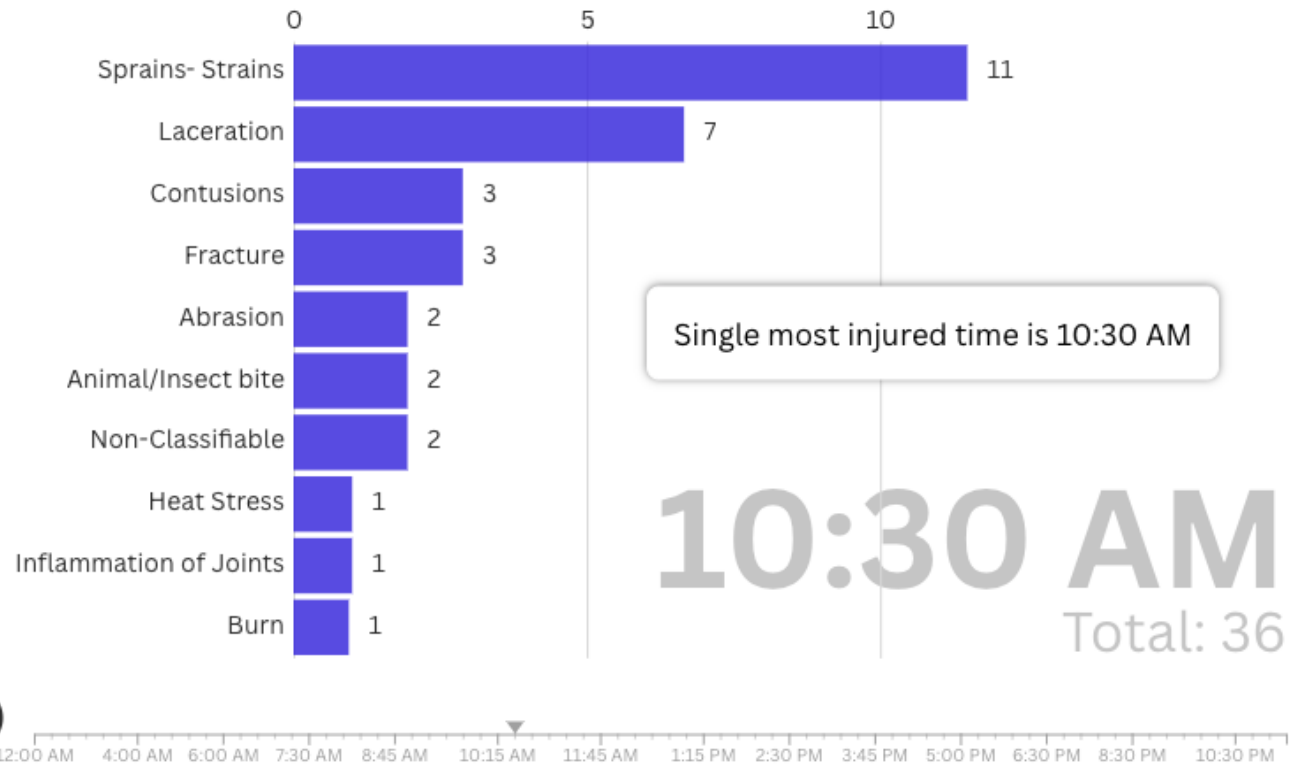
Head injuries are exclusive to men, ankle injuries are exclusive to women.



Running/animated Bar Chart

Injuries Throughout Time of Day

Injury Type



Time of Day

Source: Generic Utility Company Injury Data

Link to Flourish: <https://public.flourish.studio/visualisation/20649052/>

How Injuries Vary Throughout Day

Most common injury throughout the day is sprains/strains and occurs almost every 15 minutes.

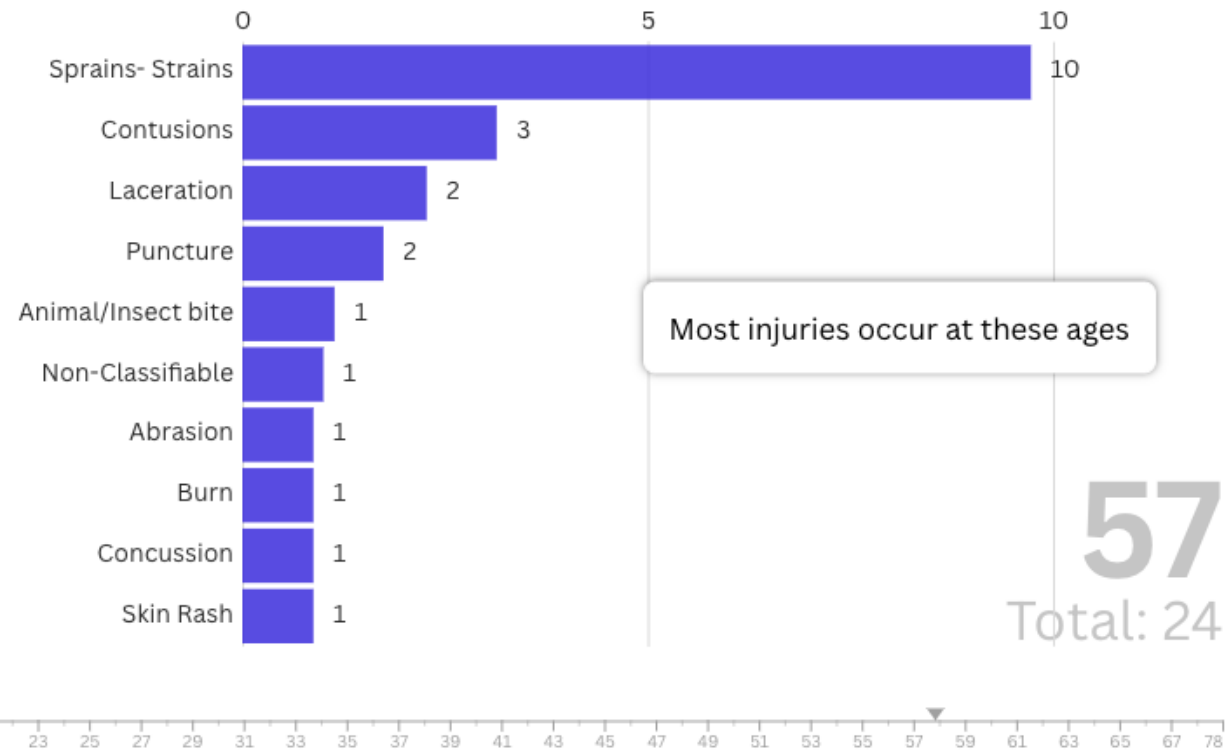
Injuries mainly occur from 8:00 AM - 3:00 PM which is a majority of the workday.

Many more injuries happen before workday compared to after, possibly related to morning commutes.

Running/animated Bar Chart

Injuries as Male Workers Age

Injury Type



Age in Years

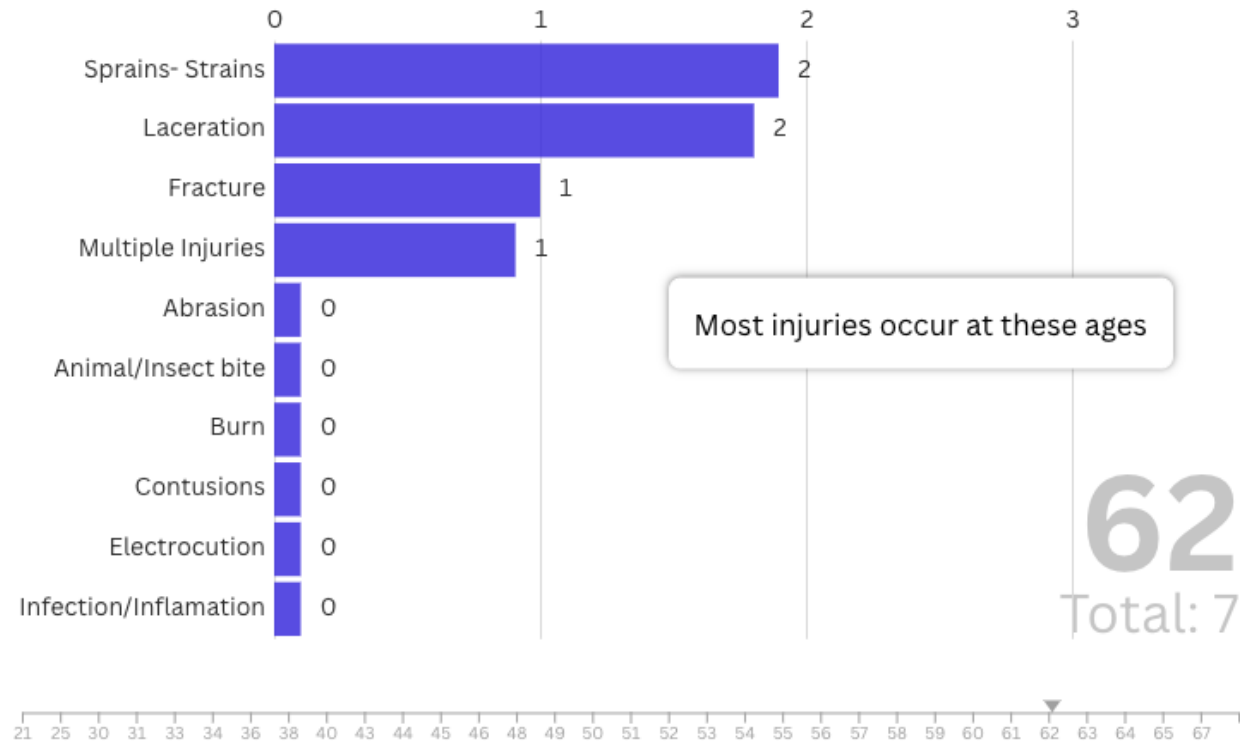
Source: Generic Utility Company Injury Data

Link to Flourish: <https://public.flourish.studio/visualisation/20662991/>

Running/animated Bar Chart

Injuries as Female Workers Age

Injury Type



Age in Years

Source: Generic Utility Company Injury Data

Link to Flourish: <https://public.flourish.studio/visualisation/20663092/>

How Injuries Change as Work Force Ages

Most common injury is sprains/strains and occurs at almost every age.

Injuries occur most often during late 50s to early 60s. They begin increasing significantly as the worker reaches age 50.

Female injuries end after age 69. Male injuries end after age 78.

Appendix

Data Cleaning

- Started w 656 rows, 15 columns
- Deleted two rows due to missing Nature of Injury and Cause of Injury data
- Moving forward with n = 654
- Shortened in Nature of Injury
 - Sprains-Strains to Sprains/Strains
 - Electrical Shock/ Electrocution to Electrocution
 - Cumulative Trauma Disorder (repetitive stress disorder) to Cumulative Trauma Disorder
 - Inflammation of Joints etc. to Inflammation of Joints
 - Combined chemical, electrical, heat/thermal burns into Burns
- Changed Net Paid to USD format
- Combined data from 10:07 AM with 10:00 AM in Time of Injury

Changes to Part of Body Affected

- Consolidated the following
 - Back = back (lower) + back (upper)
 - Legs = legs + calf + thigh
 - Feet = feet + toes
- Arms changed to Arm

PivotTable of values before
combining (combined already
small groups)



Row Labels	Count of Injury Data
Hand	100
Back - lower	75
Knees	65
Shoulder	56
Head	47
Arm	42
Ankle	36
Multiple Body Parts	32
Legs	31
Eyes	24
Face	24
Feet	21
Neck	21
Wrist	12
Ribs	11
Abdomen	10
Elbow	8
Lungs/Respiratory System	8
Back - upper	8
Hip/Buttock	8
Chest	7
Groin	3
Calf	3
Toes	2
Thigh	2
(blank)	
Grand Total	656

Additional Tableau Visualizations

Injury Types by Men and Women

Part of Body Affected	Claimant Gender	
	M	F
Hand	88	
Back	76	
Knees	53	
Shoulder	47	
Head	42	
Arm	36	
Legs	32	
Ankle	25	
Multiple Body Parts	24	
Eyes	23	
Face	21	
Neck	19	
Feet	19	
Wrist	9	
Ribs	9	
Abdomen	9	
Lungs/Respiratory System	8	
Elbow	8	
Chest	7	
Hip/Buttock	5	
Groin	3	

Injury Types by Men and Women

Part of Body Affected	Claimant Gender	
	F	M
Hand	12	
Knees	11	
Ankle	11	
Shoulder	9	
Multiple Body Parts	8	
Back	7	
Arm	6	
Head	5	
Legs	4	
Wrist	3	
Hip/Buttock	3	
Feet	3	
Face	3	
Ribs	2	
Neck	2	
Eyes	1	
Abdomen	1	

Top 3 Injury Types by Total Cost

Nature of Injury	
Sprains- Strains	\$6,116,722.92
Non-Classifiable	\$819,337.15
Fracture	\$565,542.67
Contusions	\$509,404.11
Inflammation of Joints- et..	\$336,703.86
Multiple Injuries	\$313,133.32
Tear	\$260,474.26
Abrasion	\$234,104.73
Laceration	\$200,568.42
Concussion	\$186,816.37
Dislocation	\$131,112.87
Crushing	\$122,154.39
Hernia- Rupture	\$67,769.69
Infection/Inflammation	\$35,741.61
Animal/Insect bite	\$17,734.89
Burn- Chemical	\$14,252.75
Burn- Electrical	\$9,223.90
Puncture	\$7,919.92
Heat Stress	\$7,041.49
Electrical Shock/ Electroc..	\$4,943.40
Skin Rash	\$3,090.86
Burn- Heat/Thermal	\$2,701.81
Hearing Loss or Impairme..	\$2,207.92
Null	\$1,814.57
Poisoning	\$1,779.44
Cumulative Trauma Disor..	\$1,058.84
Asphyxiation	\$314.16

PivotTable of Injury Type by Time of Day (truncated)

Count of Nature of Injury	Column Labels																						
Row Labels	1:00 AM	1:00 PM	1:15 PM	1:30 PM	1:45 PM	10:00 AM	10:00 PM	10:07 AM	10:15 AM	10:30 AM	10:30 PM	10:45 AM	10:45 PM	11:00 AM	11:00 PM	11:15 AM	11:30 AM	11:45 AM	11:45 PM	12:00 AM	12:00 PM	12:15 AM	
Abrasion		2		1		1	1		1	2	1	1				1	1		1	1		1	
Animal/Insect bite		1			1	1				2		1		2			2					3	
Asphyxiation																							
Burn							1			1							1						
Concussion										1				2									
Contusions			1		1	6				3	1	1		4		2	2	1			1	1	
Crushing			1									1					1				1		
Cumulative Trauma Disorder (repetitive stress disorder)																							
Dislocation						1																	
Electrical Shock/ Electrocutation										1	1												
Fracture		1		1		2			2	3	1	1		1				2				1	
Hearing Loss or Impairment						1																1	
Heat Stress										1		1											
Hernia- Rupture				1											1								
Infection/Inflammation		1							1														
Inflammation of Joints- etc.		1								1		1		1				1	1				
Laceration		3	2	2		6			2	7		1	1	4		2	3	1				3	
Multiple Injuries												1						1					
Non-Classifiable		2	3	2	3	1			2	2		1				1	3					1	
Poisoning																						1	
Puncture		1												1				1					
Skin Rash									1	1									1			5	
Sprains- Strains		1	11	1	9	5	10	2	1	5	12	1	3	2	10	2	3	6	6		2	11	
Tear					1		1			1							1						
(blank)																							
Grand Total		3	24	7	18	8	29	4	1	15	37	5	13	3	27	2	10	19	13	3	3	29	1

PivotTables of Injury Type by Age for Men and Women

[illegible]